Table 2. Component ablations. Retrieval contributes most to success; the cache has no detectable effect on success but saves 177 ms.

Configuration	Components				Success % ↑, ± SD	Δ vs. full pp [95% CI]	Latency ms ↓	Δ vs. full ms
	Retrieve	Verify	Route	Cache				
Full system	✓	✓	✓	✓	83.5 ±0.6	–	512	–
One component removed								
– Retrieval	✗	✓	✓	✓	75.2 ±0.7	−8.3 [−8.5, −8.0]	418	−94
– Verifier	✓	✗	✓	✓	78.8 ±0.5	−4.7 [−4.9, −4.5]	435	−77
– Routing	✓	✓	✗	✓	80.4 ±0.5	−3.1 [−3.3, −2.9]	566	+54
– Cache	✓	✓	✓	✗	83.5 ±0.6	0.0 [−0.1, 0.1]	689	+177
Two components removed								
– Retrieval, verifier	✗	✗	✓	✓	71.3 ±0.6	−12.2 [−12.4, −11.9]	341	−171
– Routing, cache	✓	✓	✗	✗	80.3 ±0.5	−3.2 [−3.4, −3.0]	748	+236

Synthetic data. Success Δ is the paired difference from the full system on the same 8 seeds, with a percentile-bootstrap 95% CI over seeds; ± is the SD across seeds. Δ is green for an improvement, red for a loss, grey for none. Shaded: the full system, the reference for Δ. – = not applicable. ✓ kept, ✗ removed.